Linear Algebra

Math Rendering Preview

Matrices, vectors, fractions, and functions

Matrix notation	Column vectors	Aligned functions

A compact, evidence-grounded review with formulas, graph rules, problem-solving methods, common traps, and a checked answer key.

Keep course-specific claims tied to their lecture and page citations. Disclose incomplete extraction or visual coverage in the review notes.

Matrices And Vectors

Compact Matrix Input

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 0 \end{bmatrix}$$

$$CA = \begin{bmatrix} 4 & 1 \\ 2 & 2 \\ 4 & 1 \end{bmatrix}$$

$$Ax = \begin{bmatrix} 3 \\ 0 \\ 3 \end{bmatrix}$$

Inverses And Fractions

$$\begin{bmatrix} 3 & 1 \\ -1 & -3 \end{bmatrix}^{-1} = \begin{bmatrix} \frac{3}{8} & \frac{1}{8} \\ -\frac{1}{8} & -\frac{3}{8} \end{bmatrix}$$

$$P = \begin{bmatrix} -\frac{1}{2} & -1 & \frac{1}{2} \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

Vector Spaces

$$\mathrm{Col}(A) = \mathrm{span} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$$

$$\mathrm{Null}(A) = \mathrm{span} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$$\mathrm{rank}(A) + \mathrm{nullity}(A) = 3$$

LaTeX Matrix Environments

$$B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$C = \begin{pmatrix} x_1 & x_2 \\ y_1 & y_2 \end{pmatrix}$$

$$\det(C) = \begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix}$$

Standard Mathematical Functions

$$\det(A) \neq 0$$

$$\lambda^2 - 3\lambda + 2 = 0$$

$$x = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{i=1}^{n} a_i x_i = b$$

$$\int_0^1 x^2 dx = \frac{1}{3}$$